Employee Attrition Prediction using Machine Learning

Executive Summary for HR Director

Project Overview

Employee turnover is one of the biggest challenges for organizations because replacing experienced employees increases recruitment costs, training expenses, and productivity loss. The objective of this project was to analyze employee data and identify employees at a higher risk of leaving the company so HR teams can take proactive retention actions.

Key Findings

- Approximately 16% of employees left the company.
- The Sales Department showed the highest attrition.
- Sales Representatives and Laboratory Technicians experienced the highest turnover.
- Employees working overtime were significantly more likely to leave.
- Lower monthly income was associated with higher attrition, although salary was not the only contributing factor.
- Most resignations occurred during the first few years of employment.

Business Impact

These insights help HR identify employees who may be considering leaving before they resign, enabling targeted retention strategies, reduced hiring costs, and improved workforce stability.

Recommendations

1. Strengthen onboarding, mentoring, and career development during employees' first three years.
2. Improve work-life balance by reducing excessive overtime and promoting flexible work policies.
3. Focus retention efforts on high-risk departments and job roles.
4. Regularly review compensation and employee recognition programs.

Project Outcome

Three machine learning models were developed and compared. The best-performing model successfully identified employees at risk of attrition and can support HR teams in making data-driven retention decisions.

Limitations

The model is based on historical employee data and cannot capture personal circumstances, career aspirations, or external job opportunities. It should be used as a decision-support tool rather than replacing HR judgment.